ANIRUDH SARVE

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate specializing in Machine Learning, Computer Vision, Deep Sensor Fusion, and High-Performance Distributed Systems. Proven track record in formulating physics-constrained Kalman filters, cross-modal remote sensing reconstruction networks, and edge neural runtimes (ONNX, WebAssembly, ARM Cortex-A) delivering mathematically guaranteed stability and sub-3 ms latency budgets. Recognized as Lead Architect and Winner of the ISRO Smart India Hackathon (SIH 2026), with hands-on experience bridging deep research theory with scalable production infrastructure.

EDUCATION

Vidyalankar Institute of Technology (VIT), Mumbai — **B.E. in Computer Engineering** **Expected May 2028**

Cumulative GPA: 8.90 / 10.0 | Affiliated with University of Mumbai | **Class XII (HSC): 92.60%** (Distinction)

Core Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Vision, Sensor Fusion, Operating Systems, Database Management Systems, Computer Networks, Linear Algebra & Probability.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript (Node.js), SQL, Go (Basics), HTML5/CSS3.

AI & Machine Learning: PyTorch, TensorFlow, ONNX, OpenCV, Faster-Whisper, Scikit-learn, 15-State ESKF, Sensor Fusion, ResU-Net, Cross-Modal Attention, LangGraph, Model Context Protocol (MCP).

Web & Distributed Systems: Next.js 16, React 19, FastAPI, Express.js, WebAssembly (WASM), WebGL, WebSockets, Docker, Linux / POSIX Sockets, Nginx, Git & GitHub.

Databases, Vector & Cloud: PostgreSQL, MongoDB (2dsphere), Upstash Redis (Distributed Locks), Qdrant Vector DB, SQLite (FTS5 BM25), AWS, Google Cloud Platform (GCP).

PROFESSIONAL EXPERIENCE

NxtChaptr — **Software Developer (Edge AI & Full-Stack Infrastructure)** **Nov 2024 – Aug 2025**

- Architected and deployed client-side edge computer vision using a U²-Net model via **WebAssembly/WebGL**, processing product image photo-matting in **<3.5 seconds** and eliminating recurring cloud GPU server costs (~$0.02/upload).
- Engineered automated catalog metadata extraction microservices using **Google Gemini Vision**, reducing merchant listing creation latency from 8 minutes to **<30 seconds**.
- Integrated Upstash Redis atomic distributed locks (SET NX EX) and Ably WebSockets to completely eliminate race-condition overselling across concurrent 1-of-1 product drops.
- Built performant responsive storefront interfaces with Next.js 16 and TypeScript, achieving sub-second Largest Contentful Paint (LCP) scores.

Wroomie — **Software Developer (Backend & Applied ML)** **Nov 2023 – Jan 2024**

- Developed multi-attribute compatibility scoring algorithms clustering user lifestyle preferences across 4 behavioral vectors, achieving **94% match satisfaction**.
- Optimized MongoDB geospatial room discovery queries with 2dsphere indexing, decreasing query latency by **35%** across high-density university zones.
- Implemented JWT-based secure authentication, input sanitization, and rate-limiting middlewares across RESTful Express service endpoints.

FEATURED TECHNICAL PROJECTS

Neural-Inertial Dead-Reckoning Engine with 15-State Error-State Kalman Filter (ISRO SIH PS 26168)

Technologies: PyTorch, 15-State ESKF, 5-Head Temporal Attention GRU, ONNX, C++, ARM Cortex-A, ROS2 Humble

- Engineered a hybrid dead-reckoning engine holding vehicle position drift to **≤ 1.85 m** across 120s GPS blackouts (a **98.4% drift reduction** versus standard EKF baselines).
- Fused 10 Hz raw IMU mechanization with a continuous-time 15-state ESKF, enforcing physical Non-Holonomic Constraints (NHC) and deep zero-velocity updates (ZUPT).
- Compiled model to a compact **24.6 KB ONNX graph** running at sub-3 ms CPU latency on ARM Cortex-A ECUs with zero cloud dependencies.

Multimodal Optical-SAR Fusion ResU-Net with Cross-Modal Attention (ISRO LISS-IV)

Technologies: PyTorch, ResU-Net, Cross-Modal Attention, Sentinel-1 SAR (C-Band), GeoTIFF, GDAL, CUDA

- Constructed generative AI framework for cloud removal in ISRO Resourcesat-2/2A LISS-IV satellite imagery (5.0 m resolution), fusing optical and SAR radar.
- Formulated differentiable Spectral Angle Mapper (SAM) loss, reaching **32.09 dB PSNR**, **0.7167 SSIM**, and tight **6.17° SAM angle** on real validation scenes.
- Engineered geospatial raster ingestion pipeline with GDAL and CUDA acceleration, processing gigapixel multi-temporal satellite tiles in minutes.

Neuro-Symbolic Hybrid Retrieval Engine & Automated Compliance Platform (SIH PS 26108)

Technologies: React 18, TypeScript, FastAPI, SQLite FTS5 BM25, ONNX Dense Embeddings, PaddleOCR v2.9+

- Built 3-way hybrid retrieval engine (SQLite FTS5 BM25 + regex patterns + 384-dim dense embeddings) via Reciprocal Rank Fusion across 19,423 BIS standards.
- Paired zero-hallucination verification kernel with PaddleOCR pipeline achieving **99.28% OCR confidence** on complex multi-column tender BoQ tables.

Sub-150ms Vector Graphics Stream PDF Compiler & Model Context Protocol (MCP) Server Monorepo

Technologies: TypeScript, Turborepo, Rust Canvas Bindings, Zod, MCP Protocol, Node.js

- Engineered vector PDF streaming engine compiling live schemas in **<150 ms (<40 MB RAM)** with zero heavy browser subprocess dependencies.
- Built native Model Context Protocol (MCP) server with self-healing Zod validation for autonomous AI agent tool execution and file workflows.

KEY ACHIEVEMENTS & HONORS

- **Winner & Lead Architect, ISRO Smart India Hackathon (SIH 2026):** Authored winning AI-ESKF sensor-fusion architecture for Problem Statement 26168.
- **Finalist, Indian Rover Challenge (IRC):** Developed onboard kinematic guidance telemetry and dead-reckoning estimators for Mars rover prototype.
- **Scholastic Distinction:** 8.90 / 10.0 CGPA at Vidyalankar Institute of Technology; 92.60% (Distinction) in Maharashtra State Board HSC Examination.
- **Open Source Contributor & Architect:** Published vector PDF streaming compilers, MCP toolkits, and computer vision utilities with 100+ GitHub stars.
- **Hackathon Mentor:** Guided junior engineering teams in computer vision model deployment, ROS2 simulation, and web application architectures.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Autonomous Systems Subsystem Lead, VIT Robotics Club:** Led telemetry and ROS2 navigation stack development for collegiate robotics competitions.
- **Technical Workshop Speaker:** Conducted hands-on technical sessions on Edge AI optimization, WebAssembly, and modern full-stack development for 200+ students.

HOBBIES & PERSONAL INTERESTS

Robotics & Hardware: Embedded ROS2/Micro-ROS tinkering, custom PCB design, autonomous ground rovers.

Astronomy & Space Tech: Satellite orbital tracking, ISRO telemetry analysis, deep-sky astrophotography.

Competitive Chess: Positional strategy and tactical opening theory (Rapid & Blitz player).

Technical Writing: Authoring engineering deep-dives on systems architecture, sensor fusion, and AI agent frameworks.